



6 A.

Data Protection & Integrity.

In Operating Systems Data is the most important part. data is used to store information and share information.

For that data needs to be stored safely and securely. As for that Data protection and integrity are used for.

Data protection → Data protection is that for the data is protected. Only authorised access people can access the data. This keeps the data protected.

Data Integrity → Data Integrity is that for the data has integrity. The data is untampered and has no loss in it is called integrity.

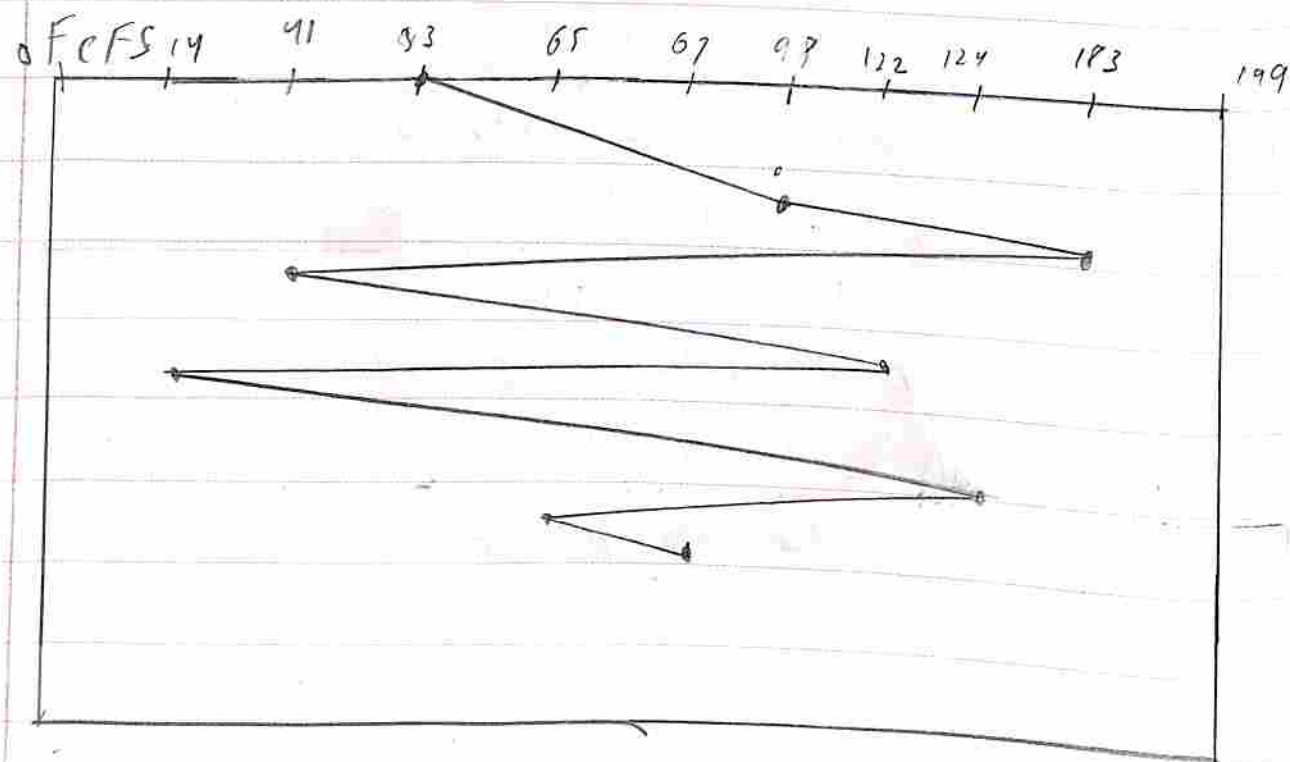
Data protection and integrity are important for data to keep safe and secure.

B)
Disk Scheduling

98, 183, 41, 122, 14, 124, 65, 67. (0-199)

Head = 53

0, 14, 41, 53, 65, 67, 98, 122, 124, 183, 199



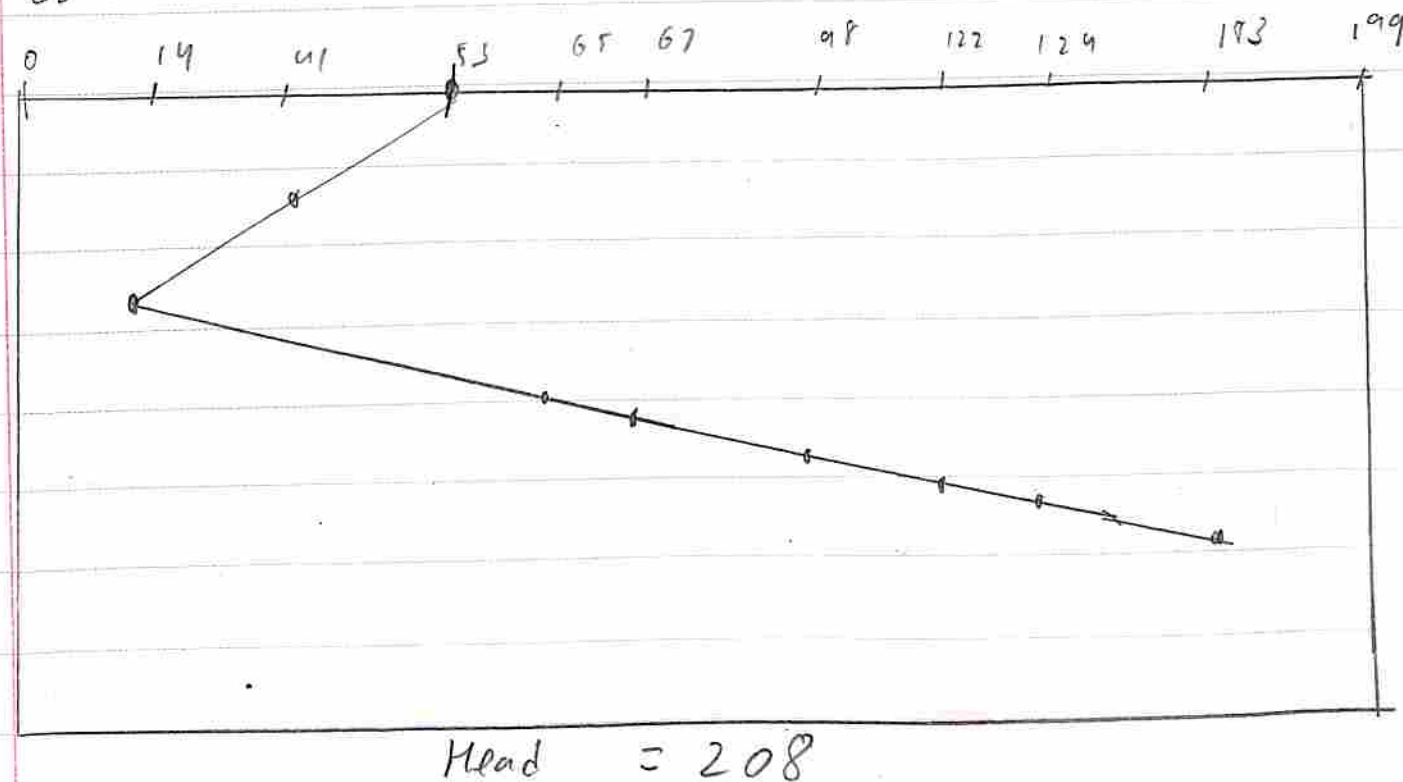
$$95 + 83 + 192 + 81 + 108 + 110 + 54 + 2$$

$$\text{Head move} = 627$$

$$12 + 27 + 51 + 2 + 31 + 24 + 2 + 59$$

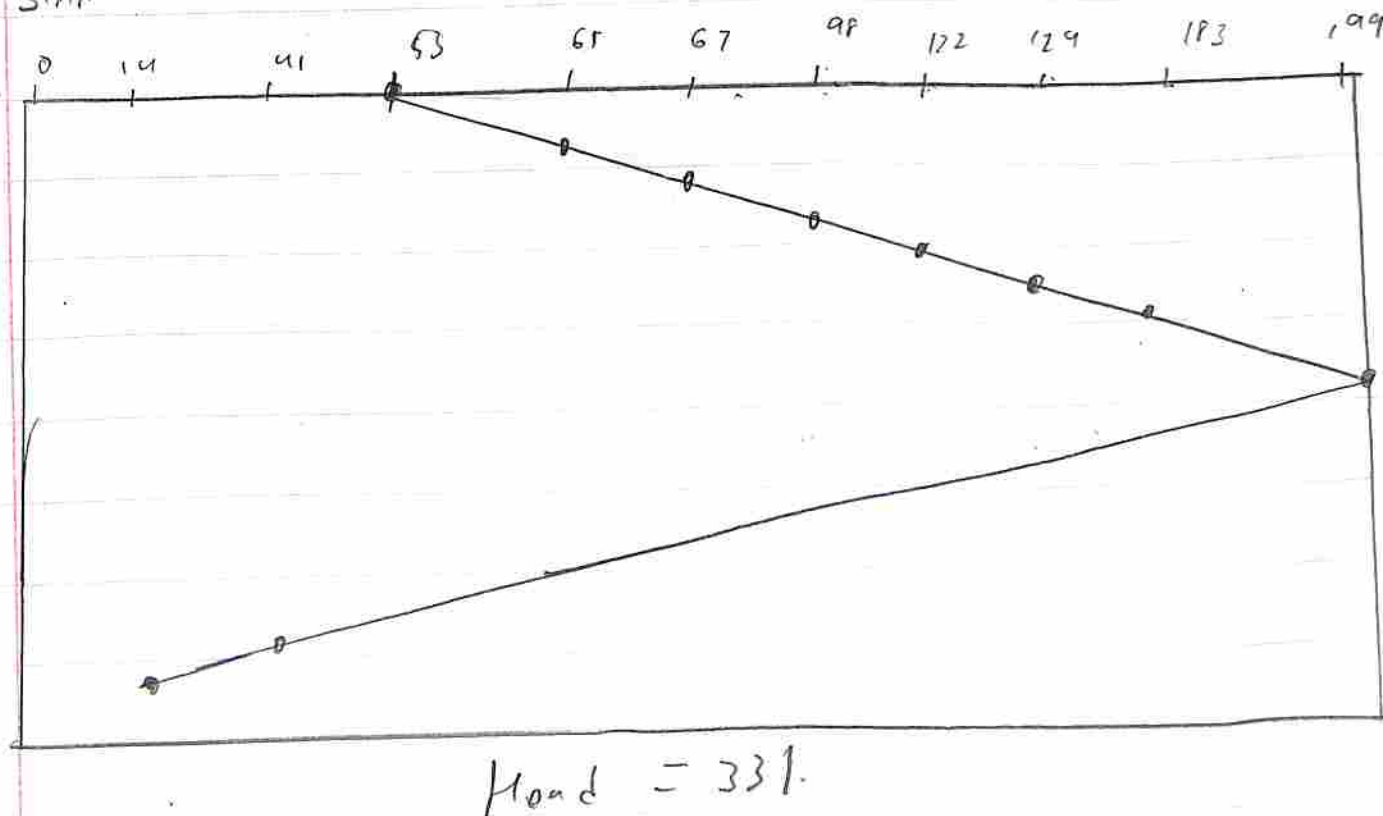
$$12/2/31/29/2/59/16/158/27$$

SSTF



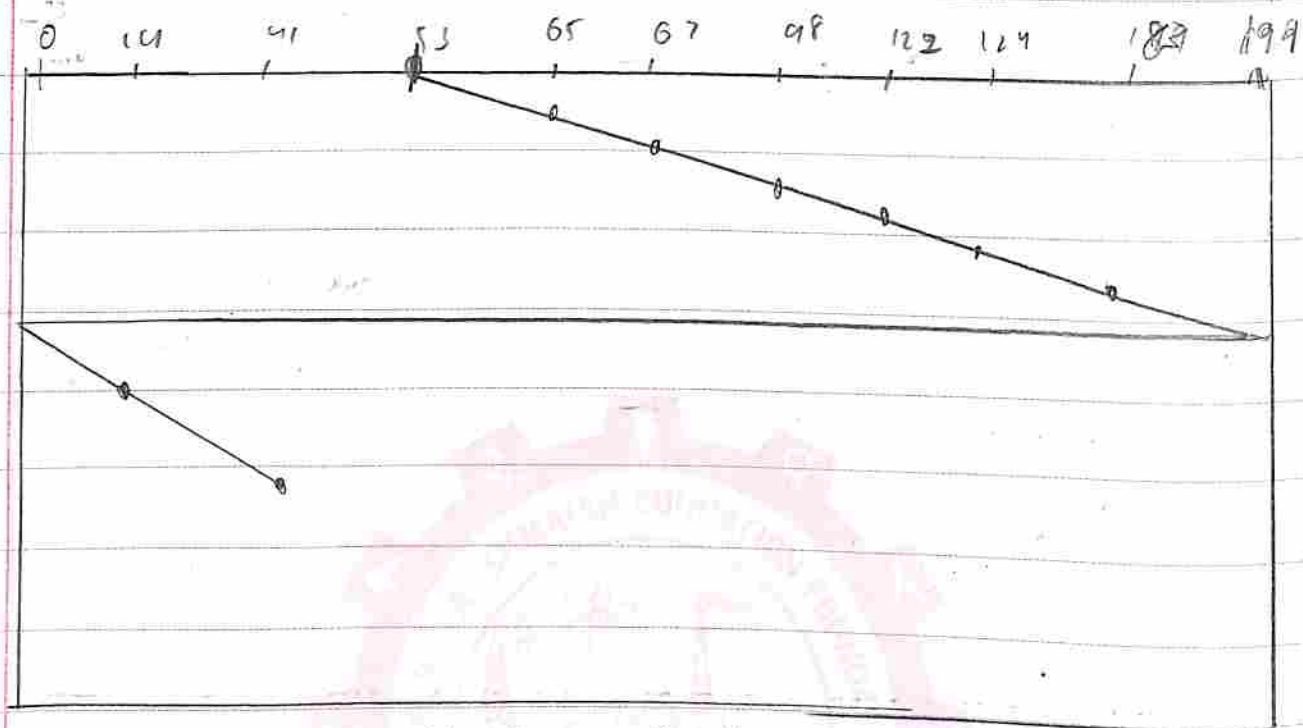
$$\text{Head} = 208$$

SCAN



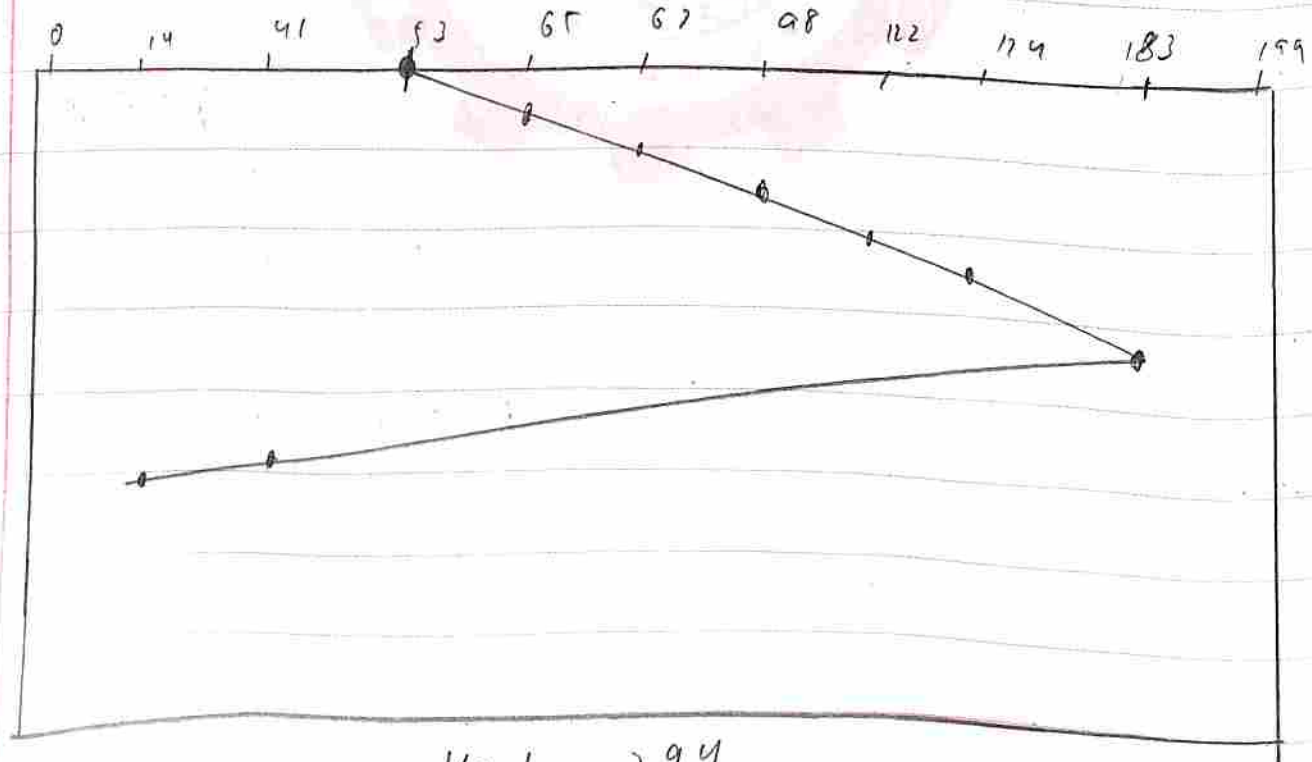
$$\text{Head} = 331$$

C-SCAN



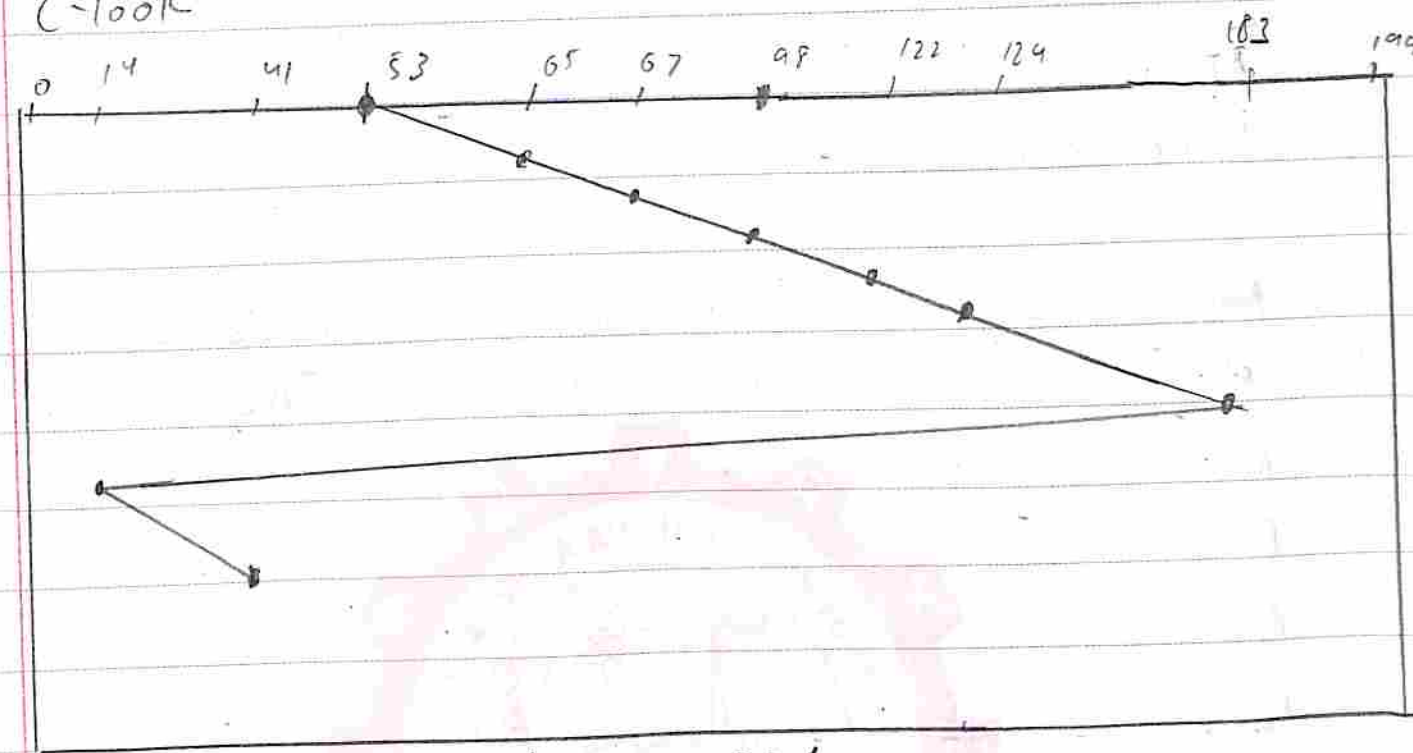
Head = 381

LOOK



Head = 294

C-look



Head = 321

$(7/2/31/29/2/59) \rightarrow 125$
 $\rightarrow (16/199/19/27) \rightarrow 256$
 $\rightarrow (142/27/169)$
 $\rightarrow (169/27) 196$

3 B

Banker's Algorithm.

Process	Allocation	max	need	Available
P	ABC	ABC	ABC	ABC
P ₀	1 1 2	4 3 3	3 2 1	2 1 0
P ₁	2 1 2	3 2 2	1 1 0	4 2 2
P ₂	4 0 1	9 0 2	5 0 1	5 3 4
P ₃	0 2 0	7 5 3	7 3 3	6 4 6
P ₄	1 1 2	1 1 2	0 0 0	10 4 7
				<u>10 6 7</u>

Cycle 1.

~~Available~~

Need < available

P₀ 3 2 1 < 2 1 0 → unsafeP₁ 1 1 0 < 2 1 0 → safe ⇒ 2 1 0 + 2 1 2 = 4 2 2P₂ 5 0 1 < 4 2 2 → unsafeP₃ 7 3 3 < 4 2 2 → unsafeP₄ 0 0 0 < 4 2 2 → safe ⇒ 4 2 2 + 1 1 2 = 5 3 4

Cycle 2.

Need < available

P₀ 3 2 1 < 5 3 4 → safe ⇒ 5 3 4 + 1 1 2 = 6 4 6P₂ 5 0 1 < 6 4 6 → safe ⇒ 6 4 6 + 4 0 1 = 10 4 7P₃ 7 3 3 < 10 4 7 → safe ⇒ 10 4 7 + 0 2 0 = 10 6 7~~P₄~~

Two Cycles

C1

C2

(P₁ → P₄) → (P₀ → P₂ → P₃)A

Conditional wait/hold.

They are unable to check if job is completed or not.

With this lock s can be done. Lock → Lock the process so that the process is at the current state as it is.

2 A/

Producer Consumer,

The producer-consumer problem is a Process Synchronization problem.

Where the producer is producing and the consumer is consuming,

Both use the shared memory space and never get tangled.

When the producer is active the consumer is sleeping.

When the producer goes to sleep it wakes up the consumer and then goes to sleep.

The same will happen when the consumer is active.

B/

Multi threading

Multi-threading is a concept where multiple threads are working on a single job but separately.

A thread is a part of an execution. In multi-threading the threads need to be synchronized so that each thread does its own job and doesn't interfere with others.

Global variables are variables in a queue to rotate it. A ~~process~~ process is completed on rotation.

Global variables has 2 pins, ~~wait~~

- wait
- signal.

With these global variables they can synchronize the threads so it can function smoothly without over-clocking.

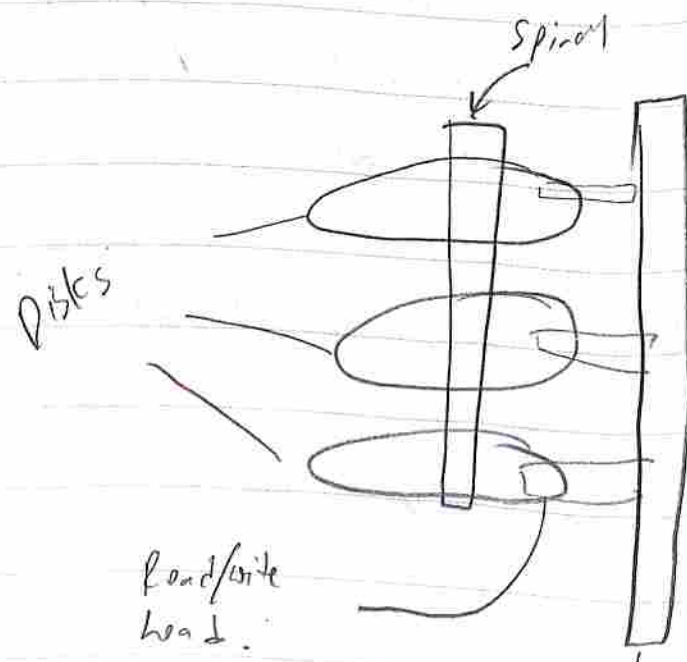
C

Network File System	Distributed File System
- File sharing system over LAN - File stored with a system in connected LAN - Fastest - Easy for process	- File sharing system over across servers - File stored on a server anywhere. - Depends on server - Hard to process

D1

Hard disk

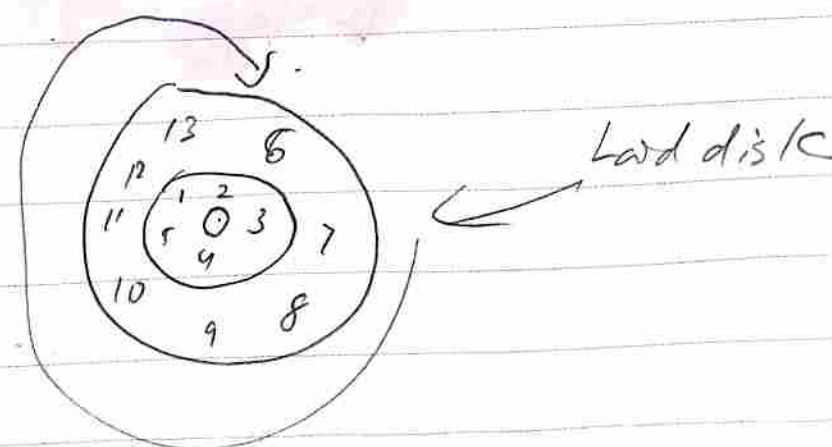
A hard disk is a combination of platter disks spinning at high speeds for data storage



4 stages

- Accelerating → start spinning the disk
- Reaching full speed in disk
- Searching if disk is readable or not
- data operations on disk

It consists of multiple disks on a spindle. Each disk has its own read/write head. Each disk has multiple tracks on it to store data.



1 A.

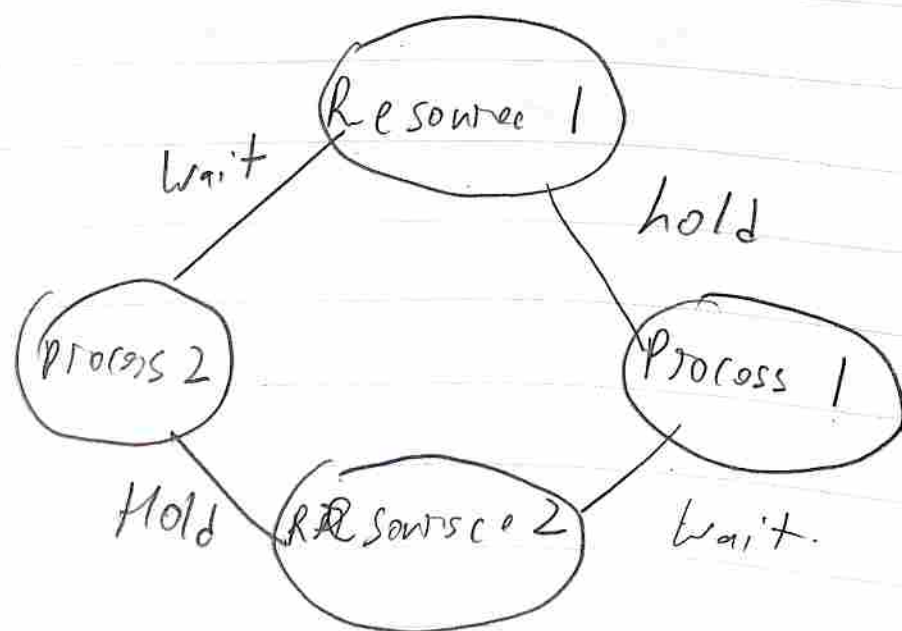
Concurrency

Concurrency is basically many to one.

When multiple people are accessing the same data, it is called Concurrency.

The same data is accessed by multiple people at the same time.

B.



A deadlock is a situation when the program doesn't move forward as each process is waiting for the other, forming a dead lock cycle.

C.

Condition Variables

A Conditional variable is a variable that informs the weather if the process is completed or not.

F.

Interrupts

An Interrupt is when the process is stopped suddenly because of an external/Internal factor.

D.

Operating System.

~~The~~ Risk $\rightarrow$ The possibility of it happening.

Threat $\rightarrow$ ~~The~~ A possible effect in system if code is run.

Attack $\rightarrow$ The OS is effected by a factor.

There are multiple ways an OS can be compromised like

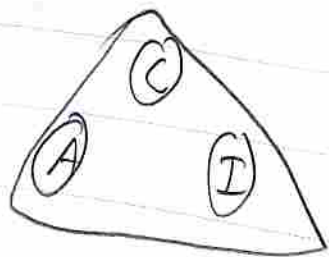
- Trojan horse → Trojan horse
- malicious code → malicious code
- Rubber ducky → Bad USB.

E

Security.

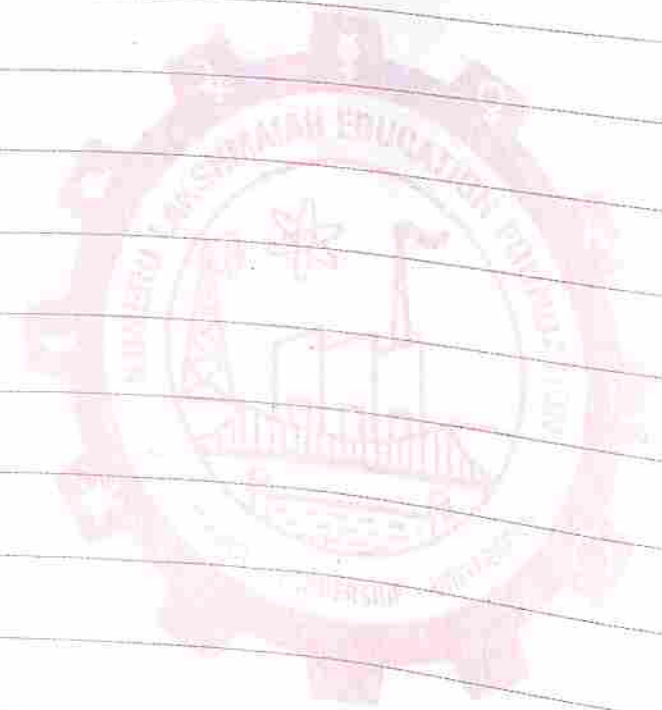
CIA triad

Confidentiality
Integrity
Authorization.



In each OS there can be security problem but each is looked after

authorization / authentication / verification / access.



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